Class Field Theory IV.3.14

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Fixed author PDF v4.03

Original statement

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THEOREM 3.14 For every finite Galois extension L/k, the map $\varphi \mapsto [A(\varphi)]$ defines an isomorphism of abelian groups $H^2(L/k) \to \mathrm{Br}(L/k)$.

Original context

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COROLLARY 3.6 Let A be a central simple algebra over k. A field L of finite degree over k splits A if and only if there exists an algebra B similar to A containing L and such that

$$[B : k] = [L : k]^2. \tag{37}$$

In particular, every subfield L of A of degree $[A : k]^{1/2}$ over k splits A.

PROOF. Suppose L splits A. Then L also splits A^{opp}, say, $A^{\mathrm{opp}} \otimes_k L = \mathrm{End}_L(V)$. This equality states that $A^{\mathrm{opp}} \otimes_k L$ is the centralizer of L in $\mathrm{End}_k(V)$, and so L is the centralizer of $A^{\mathrm{opp}} \otimes_k L$ in $\mathrm{End}_k(V)$ (see 3.1). Let B be the centralizer of A^{opp} in $\mathrm{End}_k(V)$. I claim that B satisfies the required conditions. Certainly, $B \supset L$, and Corollary 3.3 shows that B is central simple and that $B \otimes_k A^{\mathrm{opp}} \simeq \mathrm{End}_k(V)$. On tensoring both sides with A and using that $A \otimes_k A^{\mathrm{opp}}$ is a matrix algebra, we find that $B \sim A$.

For the converse, it suffices to show that L splits B. Because L is commutative, $L = L^{\mathrm{opp}} \subset B^{\mathrm{opp}}$, and because $[L : k] = \sqrt{[B : k]}$, L is equal to its centralizer in B^{opp}. Therefore the centralizer of $1 \otimes L$ in $B \otimes_k B^{\mathrm{opp}}$ is $B \otimes_k L$. When we identify $B \otimes_k B^{\mathrm{opp}}$ with $\mathrm{End}_k(B)$ (endomorphisms of B as a k-vector space—see 2.9), the centralizer of L becomes identified with $\mathrm{End}_L(B)$ (endomorphisms as an L-vector space). This completes the proof. □

Original proof and included supporting results

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Central simple algebras and 2-cocycles

Fix a finite Galois extension L of k, and let $G = \mathrm{Gal}(L/k)$. Define $\mathcal{A}(L/k)$ to be the class of central simple algebras A over k containing L and of degree $[A : k] = [L : k]^2$ (hence, L equals its centralizer in A).

Fix an $A \in \mathcal{A}(L/k)$. For every $\sigma \in G$, Corollary 2.11 of the Noether-Skolem theorem shows that there exists an element $e_\sigma \in A$ such that

$$\sigma a = e_\sigma a e_\sigma^{-1} \quad \text{for all } a \in L. \tag{38}$$

Moreover, e_σ is determined by (38) up to multiplication by an element of $L^\times$, because if f_σ has the same property, then $f_\sigma^{-1} e_\sigma$ centralizes L. Note that (38) can be written as

$$e_\sigma \cdot a = \sigma a \cdot e_\sigma \quad \text{for all } a \in L, \tag{39}$$

which says that moving e_σ past $a \in L$ replaces it with σa. Clearly $e_\sigma e_\tau$ has the property (38) for $\sigma\tau$, and so

$$e_\sigma e_\tau = \varphi(\sigma, \tau) e_{\sigma\tau} \tag{40}$$

for some $\varphi(\sigma, \tau) \in L^\times$. Note that

$$e_\rho(e_\sigma e_\tau) = e_\rho(\varphi(\sigma, \tau) e_{\sigma\tau}) = \rho\varphi(\sigma, \tau) \cdot \varphi(\rho, \sigma\tau) \cdot e_{\rho\sigma\tau}$$

and

$$(e_\rho e_\sigma) e_\tau = \varphi(\rho, \sigma) e_{\rho\sigma} e_\tau = \varphi(\rho, \sigma)\varphi(\rho\sigma, \tau) \cdot e_{\rho\sigma\tau}.$$

Therefore the associative law implies that φ is a 2-cocycle. It is even a normalized 2-cocycle if we choose $e_1 = 1$. A different choice of e_σ's leads to a cohomologous 2-cocycle, and so we have a well-defined map $A \mapsto \gamma(A) \colon \mathcal{A}(L/k) \to H^2(L/k)$.

THEOREM 3.11 The map $\gamma \colon \mathcal{A}(L/k) \to H^2(L/k)$ is surjective, and its fibres are the isomorphism classes.

We first need a lemma.

LEMMA 3.12 Let $A \in \mathcal{A}(L/k)$, and define e_σ to satisfy (38). Then the set $(e_\sigma)_{\sigma \in G}$ is a basis for A as a left vector space over L.

PROOF. Note that

$$\dim_L(A) = \frac{\dim_k(A)}{\dim_k(L)} = n,$$

and so it suffices to show that the e_σ are linearly independent. Suppose not, and let $(e_\sigma)_{\sigma \in J}$ be a maximal linearly independent set. If $\tau \notin J$, then

$$e_\tau = \sum a_\sigma e_\sigma$$

for some $a_\sigma \in L$. Let $a \in L$. When we compute $e_\tau a$ in two different ways,

$$e_\tau a = \tau a \cdot e_\tau = \sum_{\sigma \in J} \tau a \cdot a_\sigma e_\sigma,$$

$$e_\tau a = \sum_{\sigma \in J} a_\sigma e_\sigma \cdot a = \sum_{\sigma \in J} a_\sigma \cdot \sigma a \cdot e_\sigma$$

we find that $\tau a \cdot a_\sigma = \sigma a \cdot a_\sigma$ for all $\sigma \in J$. For at least one $\sigma \in J$, $a_\sigma \neq 0$, and then the equation shows that $\tau = \sigma$, contradicting the fact that $\tau \notin J$. Therefore $J = G$. □

Now A is uniquely determined by the following properties: $A \supset L$; $(e_\sigma)_{\sigma\in G}$ is a basis for A as an L-vector space; multiplication in A satisfies the equation (38) and (40).

Let $A' \in \mathcal{A}(L/k)$ and suppose that $\gamma(A) = \gamma(A')$. The condition implies that we can choose bases (e_σ) and (e'_σ) for A and A' satisfying (38) and (40) with the same 2-cocycle φ. The map $\sum a_\sigma e_\sigma \mapsto \sum a_\sigma e'_\sigma \colon A \to A'$ is an isomorphism of k-algebras.

Next suppose that A and A' are isomorphic elements of $\mathcal{A}(L/k)$. The Noether-Skolem theorem allows us to choose the isomorphism $f\colon A \to A'$ so that $f(L) = L$ and $f|_L$ is the identity map. If e_σ satisfies condition (40) for A, then $f(e_\sigma)$ satisfies (40) for A'. With the choices (e_σ) and $(f(e_\sigma))$, A and A' define the same cocycle.

These remarks show that the map $A \mapsto \gamma(A)$ defines a injection

$$\mathcal{A}(L/k)/\approx \to H^2(L/k).$$

To show that the map is surjective, we construct an inverse.

Let $\varphi\colon G\times G \to L^\times$ be a normalized 2-cocycle. Define $A(\varphi)$ to be the L-vector space with basis $(e_\sigma)_{\sigma\in G}$ endowed with the multiplication given by (38) and (40). Then e_1 is an identity element for multiplication, and the cocycle condition (exactly) shows that

$$e_\rho(e_\sigma e_\tau) = (e_\rho e_\sigma)e_\tau.$$

It follows that $A(\varphi)$ is a k-algebra. We identify L with the subfield Le_1 of $A(\varphi)$.

LEMMA 3.13 The algebra $A(\varphi)$ is central simple over K.

PROOF. Let $\alpha = \sum a_\sigma e_\sigma$ centralize L, and let $a \in L$. On comparing $a\alpha = \sum aa_\sigma \cdot e_\sigma$ with $\alpha a = \sum a_\sigma(\sigma a)\cdot e_\sigma$, we find that $a_\sigma = 0$ for $\sigma \neq 1$, and so $\alpha = a_1e_1 \in L$. Therefore, the centralizer of L in $A(\varphi)$ is L.

Let α lie in the centre of $A(\varphi)$. Then α centralizes L, and so $\alpha \in L$, say $\alpha = ae_1$, $a \in L$. On comparing $e_\sigma\cdot\alpha = (\sigma a)e_\sigma$ with $\alpha\cdot e_\sigma = ae_\sigma$, we see that $\alpha \in k$. Thus $A(\varphi)$ is central.

Let $\mathfrak{A}$ be a two-sided ideal in $A(\varphi)$; in particular, $\mathfrak{A}$ is an L-subspace of $A(\varphi)$. If $\mathfrak{A}$ contains one element e_σ, then (40) shows that it contains all, and so equals $A(\varphi)$. Suppose $\mathfrak{A} \neq 0$, and let $\alpha = \sum a_\sigma e_\sigma$ be a primordial element of $\mathfrak{A}$, with say $a_{\sigma_0} = 1$. If $a_{\sigma_1} \neq 0$, $\sigma_1 \neq \sigma_0$, then for every $a \in L$,

$$(\sigma_1 a)\cdot\alpha - \alpha\cdot a = \sum a_\sigma(\sigma_1 a - \sigma a)e_\sigma \in \mathfrak{A}.$$

If a is chosen so that $\sigma_1 a \neq \sigma_0 a$, then this element is nonzero but has fewer nonzero coefficients than α, contradicting its primordality. Therefore, $\alpha = e_{\sigma_0}$, and we have shown that $\mathfrak{A} = A(\varphi)$. □

Let φ and φ' be cohomologous 2-cocycles, say,

$$a(\sigma)\cdot\sigma a(\tau)\cdot\varphi'(\sigma,\tau) = a(\sigma\tau)\cdot\varphi(\sigma,\tau)$$

for some map $a\colon G \to L^\times$. One checks immediately that the L-linear map $A(\varphi) \to A(\varphi')$ sending e_σ to $a(\sigma)e'_\sigma$ is an isomorphism of k-algebras. Therefore $\varphi \mapsto A(\varphi)$ defines a map $H^2(L/K) \to \mathcal{A}(L/k)/\approx$, which is clearly inverse to $A \mapsto \gamma(A)$. This completes the proof of Theorem 3.11.

The algebras $A(\varphi)$ are called ***crossed-product algebra***. Before group cohomology existed, 2-cocycles $\varphi\colon G\times G \to L^\times$ were called ***factor sets*** (and sometimes still are).

To show that this map is bijective, it suffices (after Theorem 3.11) to show that the map $A \mapsto [A]\colon \mathcal{A}(L/k)/\approx \to \mathrm{Br}(L/k)$ is bijective.

If A and A' are similar central simple algebras over k, then (see 2.13) there exists a central division algebra D such that $A \sim D \sim A'$, say, $A \approx M_n(D)$, $A' \approx M_{n'}(D)$. But if $[A:k] = [A':k]$, then $n = n'$, and so $A \approx A'$. This proves that the map $\mathcal{A}(L/k)/\approx \to \mathrm{Br}(L/k)$ is injective, and 3.6 proves that it is surjective.

LEMMA 3.15 For any two 2-cocycles φ and φ', $A(\varphi+\varphi') \sim A(\varphi)\otimes_k A(\varphi')$.

PROOF. The proof is a little messy because we have to recognize $A(\varphi) \otimes_k A(\varphi')$, not as a crossed-product algebra, but as matrix algebra over a crossed-product algebra. I merely sketch the proof.[4]

Set $A = A(\varphi)$, $B = A(\varphi')$, and $C = A(\varphi + \varphi')$. Regard A and B as left L-modules (using left multiplication), and define

$$V = A \otimes_L B.$$

Concretely, V is the largest quotient space of $A \otimes_k B$ such that

$$la \otimes_L b = a \otimes_L lb$$

holds for all $a \in A$, $b \in B$, $l \in L$.

The k-vector space V has a unique right $A \otimes_k B$-module structure such that

$$(a' \otimes_L b')(a \otimes_k b) = a'a \otimes_L b'b\,, \quad \text{all } a', a \in A\,,\ b', b \in B,$$

and a unique left C-module structure such that

$$(le''_\sigma)(a \otimes_L b) = le_\sigma a \otimes_L e'_\sigma b\,, \quad \text{all } l \in L,\ \sigma \in G,\ a \in A,\ b \in B.$$

Here (e_σ), (e'_σ), and (e''_σ) are the standard bases for $A = A(\varphi)$, $B = A(\varphi')$, and $C = A(\varphi + \varphi')$ respectively.

The two actions commute, and so the right action of $A \otimes_k B$ on V defines a homomorphism of k-algebras

$$f\colon (A \otimes_k B)^{\text{opp}} \to \text{End}_C(V).$$

This homomorphism is injective because $A \otimes_k B$ (and hence its opposite) is simple. Since both $(A \otimes_k B)^{\text{opp}}$ and $\text{End}_C(V)$ have degree n^4 over k, where $n = [L : k]$, f is an isomorphism. As we showed in (1.21), any two modules over a simple ring of the same k-dimension are isomorphic, and it follows that $V \approx C^n$ as a C-module. Hence

$$\text{End}_C(V) \approx \text{End}_C(C^n) = M_n(C^{\text{opp}}),$$

and on composing this isomorphism with f we obtain an isomorphism of k-algebras

$$(A \otimes_k B)^{\text{opp}} \to M_n(C)^{\text{opp}}.$$

The same map can be interpreted as an isomorphism

$$A \otimes_k B \to M_n(C).\square$$

[4]See Blanchard 1972, pp. 94–95, or Farb and Dennis 1993, p. 126–128, for the details.